

Linear Regression

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Overview

Estimator: $\hat{\beta} = (X^\top X)^{-1} X^\top y$.

Geometry: $\hat{y} = Hy = \text{proj}_{\text{Col}(X)}(y)$.

Inference: $\hat{\beta} \mid X \sim N(\beta, \sigma^2 (X^\top X)^{-1})$.

Shapes: $X \in \mathbb{R}^{n \times p}$, $\beta \in \mathbb{R}^p$, $y \in \mathbb{R}^n$, $X^\top X \in \mathbb{R}^{p \times p}$, $X^\top y \in \mathbb{R}^p$.

Linear regression is about finding a linear relationship between inputs and an output.

Linearity and convexity guarantee optimal solutions, no need for gradient descent.

Projection geometry solves the fitting problem, and statistics quantifies its reliability.

Prediction on unseen data requires balance of interpretability and generalizability.

Simple Linear Regression

The model is $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, where Y_i is the response, x_i is the predictor, β_0 is the intercept, β_1 is the slope, and ε_i is the random error.

Equivalently, if we assume normal errors, $Y_i \mid x_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$.

The conditional mean is $\mathbb{E}[Y_i \mid x_i] = \beta_0 + \beta_1 x_i$, and the conditional variance is $\text{Var}(Y_i \mid x_i) = \sigma^2$.

Ordinary Least Squares (OLS) Estimation

We minimize $SSE(b_0, b_1) = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$. Write the residual as $e_i = y_i - b_0 - b_1 x_i$.

$\frac{\partial SSE}{\partial b_0} = -2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i)$. Setting this to zero, $\sum_{i=1}^n e_i = 0$.

Divide by n to get $\bar{y} - b_0 - b_1 \bar{x} = 0$, so $b_0 = \bar{y} - b_1 \bar{x}$. Thus the fitted line always passes through $(\bar{x}, \bar{y})$.

$\frac{\partial SSE}{\partial b_1} = -2 \sum_{i=1}^n x_i (y_i - b_0 - b_1 x_i)$. Setting this to zero, $\sum_{i=1}^n x_i e_i = 0$.

Substitute with $b_0 = \bar{y} - b_1 \bar{x}$, $\sum_{i=1}^n x_i (y_i - (\bar{y} - b_1 \bar{x}) - b_1 x_i) = 0$,

$\sum_{i=1}^n x_i [(y_i - \bar{y}) - b_1 (x_i - \bar{x})] = 0$, $\sum_{i=1}^n x_i (y_i - \bar{y}) - b_1 \sum_{i=1}^n x_i (x_i - \bar{x}) = 0$.

We now rewrite each sum by centering x_i . Because $\sum_{i=1}^n (x_i - \bar{x}) = 0$,

$\sum_{i=1}^n x_i (x_i - \bar{x}) = \sum_{i=1}^n (x_i - \bar{x} + \bar{x})(x_i - \bar{x}) = \sum_{i=1}^n (x_i - \bar{x})^2 + \bar{x} \sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n (x_i - \bar{x})^2$.

Similarly, $\sum_{i=1}^n x_i (y_i - \bar{y}) = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) + \bar{x} \sum_{i=1}^n (y_i - \bar{y}) = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$.

Therefore, $\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) - b_1 \sum_{i=1}^n (x_i - \bar{x})^2 = 0$. $b_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$.

For shorthand, write $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$ and $S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$, so $b_1 = S_{xy}/S_{xx}$.

The OLS estimators are therefore $\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$.

Correlation

Write $S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2$. The sample correlation is $r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$.

By the Cauchy-Schwarz inequality, $(S_{xy})^2 \leq S_{xx}S_{yy}$, so $-1 \leq r \leq 1$.

Equality holds iff the centered vectors are proportional: $y_i - \bar{y} = c(x_i - \bar{x})$ for all i and some constant c .

Then the points lie exactly on the line $y = \bar{y} + c(x - \bar{x})$, and $r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{c S_{xx}}{|c| S_{xx}} = \text{sign}(c) = \pm 1$.

Thus $r = 1$ when $c > 0$ (perfect positive linear fit), $r = -1$ when $c < 0$ (perfect negative linear fit), and $r = 0$ when $S_{xy} = 0$ (no linear association).

Values in between measure strength, closer to ± 1 is a tighter linear pattern, closer to 0 is a weaker one.

The sign of r is the direction, $|r|$ is the strength.

Correlation only captures linear association, so r near 0 does not rule out a strong nonlinear pattern.

The sample standard deviations are $s_x = \sqrt{S_{xx}/(n-1)}$ and $s_y = \sqrt{S_{yy}/(n-1)}$, so $\frac{s_y}{s_x} = \sqrt{\frac{S_{yy}}{S_{xx}}}$.

Therefore, $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = r \frac{s_y}{s_x}$.

The slope depends on the strength and direction of linear association through r , and on the scales of y and x through s_y/s_x . Correlation is unitless. Slope has units of $Y/\text{units of } X$.

Multiple Linear Regression

Suppose we have n observations and p parameters.

The model is $y = X\beta + \varepsilon$, where:

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n, \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1,p-1} \\ 1 & x_{21} & x_{22} & \cdots & x_{2,p-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{n,p-1} \end{bmatrix} \in \mathbb{R}^{n \times p}, \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix} \in \mathbb{R}^n, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{p-1} \end{bmatrix} \in \mathbb{R}^p.$$

The first column of X is usually a column of ones to represent the intercept.

For observation i , the model says $Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_{p-1} x_{i,p-1} + \varepsilon_i$.

Compactly: $Y_i = x_i^T \beta + \varepsilon_i$, where x_i^T is row i of the design matrix.

Multiple linear regression is linear in the parameters β , not necessarily linear in the original raw variables.

For example, $Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon_i$ is still linear regression because the parameters enter linearly.

OLS Estimation

We minimize the residual sum of squares: $SSE(\beta) = \sum_{i=1}^n (y_i - x_i^T \beta)^2$.

In matrix form: $SSE(\beta) = (y - X\beta)^T (y - X\beta)$. Equivalently, $SSE(\beta) = \|y - X\beta\|_2^2$.

$$SSE(\beta) = y^T y - y^T X\beta - \beta^T X^T y + \beta^T X^T X \beta.$$

Since $y^T X\beta$ is a scalar, $y^T X\beta = \beta^T X^T y$. Therefore, $SSE(\beta) = y^T y - 2\beta^T X^T y + \beta^T X^T X \beta$.

$$\nabla_{\beta} SSE(\beta) = -2X^T y + 2X^T X \beta = 0 \rightarrow X^T X \hat{\beta} = X^T y.$$

If $X^T X$ is invertible, then $\hat{\beta} = (X^T X)^{-1} X^T y$.

This requires that the columns of X be linearly independent, $\text{rank}(X) = p$.

Projection

The fitted vector is $\hat{y} = X\hat{\beta} = X(X^T X)^{-1} X^T y = Hy$, defining the hat matrix $H = X(X^T X)^{-1} X^T$.

The residual is $e = y - \hat{y} = (I - H)y$. Define $M = I - H$. Then $e = My$. So H produces fitted values and M produces residuals.

For a linear map $P: \mathbb{R}^n \rightarrow \mathbb{R}^n$, $\text{Im}(P) = \{Pv : v \in \mathbb{R}^n\}$ is what P can output, and $\ker(P) = \{v : Pv = 0\}$ is what P annihilates.

If P is an orthogonal projection, then equivalently $P^T = P$ and $P^2 = P$, and we get the direct sum

$$\mathbb{R}^n = \text{Im}(P) \oplus \ker(P) = \text{Im}(P) \oplus \text{Im}(I - P).$$

Every y splits uniquely as $y = Py + (I - P)y$ with $Py \in \text{Im}(P)$, $(I - P)y \in \text{Im}(I - P)$,

and the two pieces orthogonal, so $\|y\|^2 = \|Py\|^2 + \|(I - P)y\|^2$.

Image and kernel of H and M

Assume full column rank so $H = X(X^T X)^{-1} X^T$. Direct checks give symmetry and idempotence:

$$H^T = (X(X^T X)^{-1} X^T)^T = X((X^T X)^{-1})^T X^T = X(X^T X)^{-1} X^T = H,$$

$$H^2 = X(X^T X)^{-1} X^T X(X^T X)^{-1} X^T = X(X^T X)^{-1} X^T = H. \text{ For } M = I - H,$$

$$M^T = (I - H)^T = I - H^T = I - H = M,$$

$$M^2 = (I - H)^2 = I - 2H + H^2 = I - 2H + H = I - H = M.$$

So both H and M are orthogonal projectors, and M projects onto the complementary subspace.

$\text{Im}(H) = \text{Col}(X) = \text{Im}(X) = \{X\beta : \beta \in \mathbb{R}^p\}$, the set of vectors the model can fit exactly.

Equivalently $\ker(M) = \text{Im}(H)$: anything already in the model space is fixed by H and annihilated by M .

Indeed $HX = X$ and $MX = 0$. The complementary facts are $\text{Im}(M) = \ker(H) = \text{Col}(X)^\perp = \ker(X^T)$.

Thus OLS is simply: apply H to y . The normal equations $X^T e = 0$ restate $e \in \ker(X^T) = \text{Im}(M)$.

Rank and trace

For any orthogonal projection P , on $\text{Im}(P)$ we have $Pv = v$ (eigenvalue 1), and on $\ker(P)$ we have $Pv = 0$ (eigenvalue 0). So the only nonzero eigenvalue is 1, appearing $\dim \text{Im}(P)$ times.

Therefore $\text{rank}(P) = \dim \text{Im}(P) = \text{tr}(P)$, as the trace equals the sum of eigenvalues.

Apply this to H : under full column rank, $\text{Im}(H) = \text{Col}(X)$ has dimension p , so $\text{rank}(H) = \text{tr}(H) = p$.

$$\text{tr}(H) = \text{tr}(X(X^T X)^{-1} X^T) = \text{tr}((X^T X)^{-1} X^T X) = \text{tr}(I_p) = p.$$

$$\text{tr}(M) = \text{tr}(I_n) - \text{tr}(H) = n - p.$$

$\text{tr}(H) = p$ is the number of linearly independent parameters the fit uses (model degrees of freedom).

$\text{tr}(M) = n - p$ is the residual degrees of freedom, the same $n - p$ that appears in $s^2 = SSE/(n - p)$.

Regression and Error Sum of Squares

The projection split is $y = Hy + My = \hat{y} + e$, $\hat{y} \in \text{Im}(H)$, $e \in \text{Im}(M)$, with $\hat{y} \perp e$.

Pythagorean gives the uncentered identity $\|y\|^2 = \|\hat{y}\|^2 + \|e\|^2$, where $SSE = \|e\|^2 = \|My\|^2$.

We define SST (Total Sum of Squares), SSR (Regression Sum of Squares), and SSE (Error Sum of Squares):

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2, \quad SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2, \quad SSE = \|e\|^2.$$

Write $y_i - \bar{y} = (\hat{y}_i - \bar{y}) + e_i$ and expand:

$$SST = SSR + SSE + 2 \sum_{i=1}^n (\hat{y}_i - \bar{y})e_i = SSR + SSE + 2(\hat{y} - \bar{y}\mathbf{1})^T e.$$

The cross term is an inner product between a vector in $\text{Im}(H)$ and a vector in $\text{Im}(M)$. Precisely:

1. $e = My \in \text{Im}(M)$.
2. With an intercept, $\mathbf{1} \in \text{Col}(X) = \text{Im}(H)$, and $\hat{y} = Hy \in \text{Im}(H)$. Since $\text{Im}(H)$ is a subspace, $\hat{y} - \bar{y}\mathbf{1} \in \text{Im}(H)$. (Equivalently, the intercept forces $\sum e_i = \mathbf{1}^T e = 0$ and $\bar{\hat{y}} = \bar{y}$.)
3. Orthogonal projections onto complementary subspaces are orthogonal: $\text{Im}(H) \perp \text{Im}(M)$. Indeed, for any u, v , $(Hu)^\top (Mv) = u^\top H(I - H)v = u^\top (H - H^2)v = 0$, using $H^2 = H$ and $H^\top = H$.

Therefore $(\hat{y} - \bar{y}\mathbf{1})^\top e = 0$, the cross term vanishes, and $SST = SSR + SSE$. Here SSE is variation in $\text{Im}(M)$, and SSR is variation of the fitted values (around $\bar{y}$) inside $\text{Im}(H)$.

Inference

Projection gave $\hat{y} = Hy$, $e = My$, and $SST = SSR + SSE$, but fitting alone does not quantify uncertainty in the estimated variable $\hat{\beta}$. Here we add a probability model and derive $\mathbb{E}[\hat{\beta}]$ and $\text{Var}(\hat{\beta})$.

Assume throughout that X is treated as fixed (or we condition on X) and has full column rank, so $\hat{\beta} = (X^T X)^{-1} X^T y$ exists and is unique.

For means, variances, and later t/F statements we add: $\varepsilon \sim N(0, \sigma^2 I)$, equivalently $y \sim N(X\beta, \sigma^2 I)$. In components this means $\mathbb{E}[\varepsilon_i] = 0$, $\text{Var}(\varepsilon_i) = \sigma^2$, $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$, and joint normality.

$\hat{\beta}$ as the true β plus noise

Start from the model $y = X\beta + \varepsilon$ and the OLS formula $\hat{\beta} = (X^T X)^{-1} X^T y$.

Expand: $\hat{\beta} = (X^T X)^{-1} X^T X\beta + (X^T X)^{-1} X^T \varepsilon = \beta + (X^T X)^{-1} X^T \varepsilon$.

So every randomness in $\hat{\beta}$ comes from the linear image of ε under the fixed map $A = (X^T X)^{-1} X^T$.

Take expectations, treating X as fixed: $\mathbb{E}[\hat{\beta}] = \beta + (X^T X)^{-1} X^T \mathbb{E}[\varepsilon] = \beta$, since $\mathbb{E}[\varepsilon] = 0$.

Thus OLS is unbiased for β under zero-mean errors.

Variance and Covariance

From $\hat{\beta} = \beta + A\varepsilon$ with $A = (X^T X)^{-1} X^T$, $\text{Var}(\hat{\beta}) = \text{Var}(A\varepsilon)$.

For a fixed matrix A and a random vector with $\text{Var}(\varepsilon) = \sigma^2 I$, $\text{Var}(A\varepsilon) = A \text{Var}(\varepsilon) A^\top = \sigma^2 A A^\top$.

Compute $A A^\top = (X^T X)^{-1} X^T \cdot X ((X^T X)^{-1})^\top$. Symmetry of $X^T X$ implies symmetry of $(X^T X)^{-1}$.

$A A^\top = (X^T X)^{-1} X^T X (X^T X)^{-1} = (X^T X)^{-1}$. Therefore $\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$.

Conditional on X , one writes the same as $\text{Var}(\hat{\beta} | X) = \sigma^2 (X^T X)^{-1}$.

Everything about coefficient uncertainty is stored in $(X^T X)^{-1}$, scaled by the noise level σ^2 .

Write $C = (X^T X)^{-1}$. Then $\text{Var}(\hat{\beta}_j) = \sigma^2 C_{jj}$, $\text{Cov}(\hat{\beta}_j, \hat{\beta}_k) = \sigma^2 C_{jk}$. Large C_{jj} means an uncertain $\hat{\beta}_j$.

If $\varepsilon \sim N(0, \sigma^2 I)$, then $A\varepsilon$ is multivariate normal (linear image of a normal), and $\hat{\beta} \sim N(\beta, \sigma^2 (X^T X)^{-1})$. In particular $\hat{\beta}_j \sim N(\beta_j, \sigma^2 C_{jj})$. If σ^2 were known, $(\hat{\beta}_j - \beta_j) / (\sigma \sqrt{C_{jj}}) \sim N(0, 1)$. In practice σ^2 is unknown, so we estimate it and later replace the normal pivot by a t pivot.

Standard Deviations and Standard errors

The residual is $e = y - \hat{y} = My$ with $M = I - H$. Because $MX = 0$, $e = M(X\beta + \varepsilon) = M\varepsilon$, so the residuals are a known linear image of the errors. Then $SSE = e^T e = \varepsilon^T M \varepsilon$, a 1×1 matrix, hence a scalar equal to its own trace.

$$\mathbb{E}[\varepsilon^T M \varepsilon \mid X] = \mathbb{E}[\text{tr}(M \varepsilon \varepsilon^T) \mid X] = \text{tr}(M \mathbb{E}[\varepsilon \varepsilon^T \mid X]).$$

Now use $\mathbb{E}[\varepsilon \mid X] = 0$, $\mathbb{E}[\varepsilon \varepsilon^T \mid X] = \text{Var}(\varepsilon \mid X) + \mathbb{E}[\varepsilon \mid X] \mathbb{E}[\varepsilon \mid X]^T = \sigma^2 I + 0 = \sigma^2 I$.

Therefore $\mathbb{E}[SSE \mid X] \mathbb{E}[\varepsilon^T M \varepsilon \mid X] = \text{tr}(M \cdot \sigma^2 I) = \sigma^2 \text{tr}(M) = \sigma^2(n - p)$.

So the expected residual sum of squares is σ^2 times the residual dimension.

Therefore $s^2 = \frac{SSE}{n-p}$ is unbiased for σ^2 . Under normal errors one also has $SSE/\sigma^2 \sim \chi_{n-p}^2$, which is why s^2 connects to t and F later.

Replace σ^2 by s^2 in the variance formula: $\widehat{\text{Var}}(\hat{\beta}_j) = s^2 C_{jj}$, $SE(\hat{\beta}_j) = s \sqrt{C_{jj}}$.

So $SE(\hat{\beta}_j)$ grows when residual noise s is large, when column j has little unique variation (large C_{jj}), or when n is small. Near multicollinearity is the case where some C_{jj} become huge while $\hat{y}$ may still look fine.

t -tests for one coefficient

If σ were known, $(\hat{\beta}_j - \beta_j) / (\sigma \sqrt{C_{jj}}) \sim N(0, 1)$. With σ replaced by s , and using normal errors with $(n - p)s^2/\sigma^2 \sim \chi_{n-p}^2$, the usual normal-over-estimated-SE pivot becomes Student's t : $\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} \sim t_{n-p}$.

To test $H_0 : \beta_j = 0$ vs $H_A : \beta_j \neq 0$, use $t_j = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)} = \frac{\hat{\beta}_j}{s \sqrt{C_{jj}}}$. Under H_0 , $t_j \sim t_{n-p}$.

More generally, for $H_0 : \beta_j = a$, replace the numerator by $\hat{\beta}_j - a$.

It asks: other columns controlled, is there evidence that $\beta_j \neq 0$? It does not ask whether x_j would be useful alone, or whether the whole feature set predicts well. If x_1 and x_2 are highly correlated, each may have a weak t while they are jointly useful, and that's why we also use overall and partial F -tests.

A level $1 - \alpha$ Confidence Interval for β_j is $\hat{\beta}_j \pm t_{n-p, 1-\alpha/2} SE(\hat{\beta}_j)$. Under repeated sampling with the same X , about $1 - \alpha$ of such intervals cover the true β_j .

Prediction at a new x_0

Let $x_0 \in \mathbb{R}^p$ be a new feature vector (including a 1 for the intercept if the model has one). The fitted mean is $\hat{y}_0 = x_0^T \hat{\beta}$, estimating $\mu_0 = x_0^T \beta = \mathbb{E}[Y \mid x_0]$.

Because $\hat{y}_0 = x_0^T \hat{\beta}$, $\text{Var}(\hat{y}_0) = x_0^T \text{Var}(\hat{\beta}) x_0 = \sigma^2 x_0^T (X^T X)^{-1} x_0$. Estimate with $SE(\hat{y}_0) = s \sqrt{x_0^T (X^T X)^{-1} x_0}$. A CI for the mean response is $\hat{y}_0 \pm t_{n-p, 1-\alpha/2} SE(\hat{y}_0)$.

A new observation is $Y_{\text{new}} = x_0^\top \beta + \varepsilon_{\text{new}}$. If ε_{new} is independent of the training sample and has variance σ^2 , then $\text{Var}(Y_{\text{new}} - \hat{y}_0) = \sigma^2 + \text{Var}(\hat{y}_0) = \sigma^2 \left(1 + x_0^\top (X^\top X)^{-1} x_0\right)$. Thus $SE_{\text{pred}} = s \sqrt{1 + x_0^\top (X^\top X)^{-1} x_0}$, and a prediction interval is $\hat{y}_0 \pm t_{n-p, 1-\alpha/2} SE_{\text{pred}}$. The extra 1 is irreducible noise, and the remaining term is uncertainty in the estimated mean surface.

Overall F -test

With an intercept, the overall F -test asks whether the slopes improve on the intercept-only model.

The null is $H_0 : \beta_1 = \dots = \beta_{p-1} = 0$ (intercept free).

Using $SST = SSR + SSE$ from projection, $F = \frac{SSR/(p-1)}{SSE/(n-p)} = \frac{MSR}{MSE}$.

Under H_0 and normal errors, $F \sim F_{p-1, n-p}$. Equivalently, with $R^2 = SSR/SST$, $F = \frac{R^2/(p-1)}{(1-R^2)/(n-p)}$.

A small R^2 can still be significant if n is large, while a large R^2 is not necessarily significant if n is small and p is large. Significance is not the same as practical or out-of-sample usefulness.

Partial F -tests

Compare a reduced model M_R with p_R parameters to a full model M_F with $p_F > p_R$ parameters, where M_R is obtained by setting some coefficients in M_F to zero.

Example: full $Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$

vs reduced $Y = \beta_0 + \beta_1 x_1 + \varepsilon$. $H_0 : \beta_2 = \beta_3 = 0$.

Let SSE_R and SSE_F be the residual sums of squares. Enlarging the column space cannot increase training SSE, so $SSE_F \leq SSE_R$. The partial F statistic is $F = \frac{(SSE_R - SSE_F)/(p_F - p_R)}{SSE_F/(n - p_F)}$. Under H_0 , $F \sim F_{p_F - p_R, n - p_F}$. The numerator is extra fit per added parameter; the denominator is residual noise in the full model.

Optional Extensions

Comments

With an intercept, $R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$ is the fraction of variation around $\bar{y}$ explained by $\text{Im}(H)$. Adding columns enlarges $\text{Im}(H)$, so training SSE cannot rise and training R^2 cannot fall. High training R^2 alone does not mean good prediction. Unlike simple regression, R^2 is not one pairwise r^2 .

In $Y = \beta_0 + \beta_1 x_1 + \dots + \beta_{p-1} x_{p-1} + \varepsilon$, the coefficient β_j is the expected change in Y when x_j increases by one unit and the other predictors are held fixed. That “hold fixed” story is only clean if the other predictors can really stay fixed while x_j moves. When predictors are entangled, we need a precise version of “unique variation” (FWL below) and we meet instability (special cases next).

Ordinary OLS writes $\hat{\beta} = (X^\top X)^{-1} X^\top y$. This needs $X^\top X$ invertible, i.e. $\text{rank}(X) = p$ (full column rank). Then $\hat{\beta}$ is unique. The fitted vector $\hat{y} = Hy$ is always the unique projection onto $\text{Im}(H)$.

If one column is a linear combination of others, $\text{rank}(X) < p$ and $X^\top X$ is singular. Example $x_3 = x_1 + x_2$. Infinitely many β share the same fitted values, then individual coefficients are not identifiable.

If $p > n$, then $\text{rank}(X) \leq n < p$, so $X^\top X$ is never invertible. Many β can achieve the same minimal training error (often zero if $\text{rank}(X) = n$). Coefficients are not unique; one must pick a rule (e.g. minimum norm) to choose among them.

Near multicollinearity

OLS works by projecting y onto the column space of X . When the columns of X are independent and well separated, the coefficient vector $\hat{\beta}$ is stable. When the columns of X are exactly dependent, $\hat{\beta}$ is not uniquely identified. When the columns of X are almost dependent, $\hat{\beta}$ exists but can be very unstable.

Suppose X has full column rank, so $X^T X$ is invertible and $\hat{\beta} = (X^T X)^{-1} X^T y$.

Near multicollinearity means the columns of X are technically linearly independent, but some column is almost a linear combination of the others.

For example, if $x_2 \approx 3x_1 - 2x_3$, then the columns are not exactly dependent, but there is very little independent information in x_2 after accounting for x_1 and x_3 .

So the model may have many coefficient vectors that produce nearly the same predictions. This explains why near multicollinearity often causes: large standard errors, unstable coefficient signs, large changes in coefficients when the data change slightly, and difficulty interpreting individual coefficients.

Suppose x_1 and x_2 are centered and normalized so that $x_1^T x_1 = 1$, $x_2^T x_2 = 1$, $x_1^T x_2 = \rho$.

Ignoring the intercept for the moment, we have $X^T X = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$, $(X^T X)^{-1} = \frac{1}{1-\rho^2} \begin{bmatrix} 1 & -\rho \\ -\rho & 1 \end{bmatrix}$.

As $|\rho| \rightarrow 1$, the factor $\frac{1}{1-\rho^2}$ becomes very large.

Under the usual homoskedastic model, $\text{Var}(\hat{\beta} | X) = \sigma^2 (X^T X)^{-1}$.

This is the variance inflation idea.

Singular Value Decomposition (SVD)

Let $X \in \mathbb{R}^{n \times p}$ have rank r . The thin SVD of X is $X = U_r \Sigma_r V_r^T$, where $U_r \in \mathbb{R}^{n \times r}$, $\Sigma_r = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r)$, $V_r \in \mathbb{R}^{p \times r}$.

The columns of U_r are orthonormal vectors in observation space $\mathbb{R}^n$.

The columns of V_r are orthonormal vectors in coefficient space $\mathbb{R}^p$.

The singular values satisfy $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$.

The SVD says that X acts in three steps:

First, V_r^T rotates coefficient space into special coordinate directions.

Second, Σ_r stretches each direction by σ_j .

Third, U_r places the result into observation space.

So if σ_j is small, then the corresponding coefficient direction v_j barely changes the fitted vector $X\beta$.

This is exactly near multicollinearity.

If X has full column rank, then $r = p$ and $X^T X = V \Sigma^2 V^T$, $(X^T X)^{-1} = V \Sigma^{-2} V^T$.

Then $\hat{\beta} = (X^T X)^{-1} X^T y = V \Sigma^{-2} V^T V \Sigma U^T y = V \Sigma^{-1} U^T y$.

Equivalently, $\hat{\beta} = \sum_{j=1}^p \frac{u_j^T y}{\sigma_j} v_j$.

If σ_j is small, then the component $u_j^T y$ is divided by a small number. Small singular values amplify noise.

Moore-Penrose pseudoinverse

When $\text{rank}(X) = r < p$, the map X annihilates a nontrivial $\ker(X)$, so $X^T X$ is singular and $(X^T X)^{-1}$ does not exist. Least squares still has a clear geometric goal from projection: choose $\hat{y}$ as the orthogonal projection of y onto $\text{Col}(X)$. What fails is uniqueness of the coefficient vector β with $X\beta = \hat{y}$. We need a rule that (i) still produces that projection and (ii) picks one canonical β .

Start from SVD with $\sigma_1 \geq \dots \geq \sigma_r > 0$, and every remaining diagonal entry is 0. Define the pseudoinverse of Σ to be $\Sigma^+ \in \mathbb{R}^{p \times n}$ by the entrywise rule: put $1/\sigma_j$ where $\sigma_j > 0$, and put 0 where Σ has a 0. Then $X^+ = V\Sigma^+U^T$. Then those kernel directions stay dead. That is exactly “reciprocal or zero.”

Define $H = XX^+ = (U_r \Sigma_r V_r^T)(V_r \Sigma_r^{-1} U_r^T) = U_r \Sigma_r (V_r^T V_r) \Sigma_r^{-1} U_r^T = U_r \Sigma_r I_r \Sigma_r^{-1} U_r^T = U_r U_r^T$.

The columns of U_r are an orthonormal basis of $\text{Col}(X)$. H is the orthogonal projector onto $\text{Col}(X)$, the same H as in the full-rank case $H = X(X^T X)^{-1} X^T$, but now defined without inverting $X^T X$. The residual maker is still $M = I - H = I - XX^+$, and $\hat{y} = Hy = XX^+y = \text{proj}_{\text{Col}(X)}(y)$ is unique even when β is not.

Set $\hat{\beta}^+ = X^+y$. Then $X\hat{\beta}^+ = XX^+y = \hat{y}$, so $\hat{\beta}^+$ is a least-squares solution. All least-squares solutions are $\hat{\beta}^+ + z$, $z \in \ker(X)$, because $X(\hat{\beta}^+ + z) = X\hat{\beta}^+ + Xz = X\hat{\beta}^+$. Adding $z \in \ker(X)$ does not change fitted values; it only changes which coordinates we use for the same $\hat{y}$.

Why is $\hat{\beta}^+$ the shortest? By construction $X^+ = V_r \Sigma_r^{-1} U_r^T$, so $\hat{\beta}^+ = X^+y$ lies in $\text{Col}(V_r) = \text{Col}(X^T)$ $= \ker(X)^\perp$. For any other LS solution $\hat{\beta}^+ + z$ with $z \in \ker(X)$ we have $\hat{\beta}^+ \perp z$, hence $\|\hat{\beta}^+ + z\|_2^2 = \|\hat{\beta}^+\|_2^2 + \|z\|_2^2 \geq \|\hat{\beta}^+\|_2^2$. So among all β that realize the same optimal $\hat{y}$, the pseudoinverse picks the unique minimum-norm one. Problem solved.

Fallback cases:

- If X is square and invertible, then $r = n = p$, every $\sigma_j > 0$, and $\Sigma^+ = \Sigma^{-1}$, so $X^+ = V\Sigma^{-1}U^T = X^{-1}$. Ordinary inverse.
- If X has full column rank ($r = p \leq n$), there is no kernel in coefficient space. The unique LS solution is the OLS formula already derived, $(X^T X)^{-1} X^T y$. Matching the SVD expression $V\Sigma^{-1}U^T y$ from the previous subsection shows $X^+ = (X^T X)^{-1} X^T$. Same object: unique $\hat{\beta}$, which is automatically minimum-norm.

Standardization

Standardization changes the units of predictors.

For a predictor x_j , define $z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}$, where $\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$ and $s_j^2 = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2$.

Then z_j has sample mean 0 and sample variance 1.

If we regress y on standardized predictors, the model is

$$y_i = \gamma_0 + \gamma_1 z_{i1} + \dots + \gamma_{p-1} z_{i,p-1} + \varepsilon_i.$$

The coefficient γ_j means:

If x_j increases by one sample standard deviation, holding the other standardized predictors fixed, the predicted value of y changes by γ_j units.

In ordinary unpenalized OLS with an intercept, standardizing predictors does not change the fitted values. It only changes the coordinate system used to describe the same fitted vector. To see this, suppose $x_{ij} = \bar{x}_j + s_j z_{ij}$.

If the standardized model is $\hat{y}_i = \hat{\gamma}_0 + \sum_{j=1}^{p-1} \hat{\gamma}_j z_{ij}$,

then substituting $z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}$ gives $\hat{y}_i = \hat{\gamma}_0 + \sum_{j=1}^{p-1} \hat{\gamma}_j \frac{x_{ij} - \bar{x}_j}{s_j}$.

So in the original units, $\hat{\beta}_j = \frac{\hat{\gamma}_j}{s_j}$ and $\hat{\beta}_0 = \hat{\gamma}_0 - \sum_{j=1}^{p-1} \frac{\hat{\gamma}_j \bar{x}_j}{s_j}$.

Thus the model fit is the same, but the coefficients are expressed in different units.

Regularization

Standardization becomes much more important when we use regularization. Regularization modifies the least squares objective by penalizing coefficient size. The goal is not to minimize training error as much as possible, but to improve prediction on new data. This is a bias-variance tradeoff. OLS is unbiased under the usual assumptions, but it can have high variance when predictors are correlated or when the model is very flexible. Regularization deliberately introduces bias to reduce variance.

Ridge regression solves $\min_{\beta} \|y - X\beta\|^2 + \lambda\|\beta\|^2$, where $\lambda \geq 0$. Usually the intercept is not penalized. In practice, we often center y and standardize the non-intercept predictors, then penalize only the slopes. If X contains only the predictors being penalized and no intercept, the ridge solution is $\hat{\beta}_{ridge} = (X^T X + \lambda I)^{-1} X^T y$. When $\lambda = 0$, ridge becomes OLS if $X^T X$ is invertible. When $\lambda > 0$, the matrix $X^T X + \lambda I$ is invertible even if $X^T X$ is singular. This is one reason ridge is useful with multicollinearity.

Lasso solves $\min_{\beta} \|y - X\beta\|^2 + \lambda\|\beta\|_1$, where $\|\beta\|_1 = \sum_j |\beta_j|$. Lasso can set some coefficients exactly equal to zero. So lasso does both shrinkage and variable selection. Ridge usually shrinks coefficients but does not set them exactly to zero. Elastic net combines the two penalties: $\min_{\beta} \|y - X\beta\|^2 + \lambda_1\|\beta\|_1 + \lambda_2\|\beta\|^2$.

In practical machine learning, λ is usually chosen by validation error or cross-validation. We fit the model on training data for several candidate values of λ . Then we choose the value that performs best on held-out data. This is different from choosing the model that has the smallest training SSE. Training SSE always tends to improve when the model becomes more flexible. The point of regularization is to improve out-of-sample prediction.

Exercises

Exercise 1. Gauss-Markov theorem

Assume the classical linear model $y = X\beta + \varepsilon$, where $X \in \mathbb{R}^{n \times p}$, $\text{rank}(X) = p$, $E(\varepsilon | X) = 0$, $\text{Var}(\varepsilon | X) = \sigma^2 I_n$.

Let $c \in \mathbb{R}^p$ be fixed. We want to estimate the scalar parameter $\theta = c^T \beta$.

Consider linear estimators of the form $a^T y$, where $a \in \mathbb{R}^n$.

- (i) Find the condition on a for $a^T y$ to be unbiased for $c^T \beta$.
- (ii) Show that the OLS estimator $c^T \hat{\beta}$ can be written as an unbiased $a_0^T y$.
- (iii) Show that every unbiased $c^T \beta$ can be written as $a^T y = (a_0 + h)^T y$ where $h \in \ker(X^T)$.
- (iv) Prove that $\text{Var}(a^T y | X) = \text{Var}(a_0^T y | X) + \sigma^2 \|h\|^2$. Conclude that $c^T \hat{\beta}$ has the smallest variance among all linear unbiased estimators of $c^T \beta$.

Solution 1

(i) $a^T y = a^T X\beta + a^T \varepsilon$. $E(a^T y | X) = a^T X\beta + a^T E(\varepsilon | X) = a^T X\beta$.

$a^T X\beta = c^T \beta$ for every possible $\beta \in \mathbb{R}^p$. $a^T X = c^T$. Equivalently, $X^T a = c$.

(ii) $c^T \hat{\beta} = c^T (X^T X)^{-1} X^T y = [X(X^T X)^{-1} c]^T y$. Define $a_0 = X(X^T X)^{-1} c$. Then $c^T \hat{\beta} = a_0^T y$.

$X^T a_0 = X^T X(X^T X)^{-1} c = c$. So a_0 satisfies the unbiasedness condition from part (i).

(iii) Let $a^T y$ be any linear unbiased estimator of $c^T \beta$. Then $X^T a = c$.

From part (ii), $X^T a_0 = c$. Then, $X^T(a - a_0) = 0$, so $h = a - a_0 \in \ker(X^T)$.

Therefore $a = a_0 + h$, $h \in \ker(X^T)$.

Conversely, if $h \in \ker(X^T)$, then $X^T(a_0 + h) = X^T a_0 + X^T h = c + 0 = c$.

Thus every linear unbiased estimator is OLS plus an extra vector h that is invisible to X^T .

(iv) Since $a = a_0 + h$, we have $a^T y = a_0^T y + h^T y$.

Conditional on X , $\text{Var}(a^T y | X) = \text{Var}((a_0 + h)^T y | X)$.

Because $y = X\beta + \varepsilon$ and $X\beta$ is fixed given X , $\text{Var}(a^T y | X) = \text{Var}((a_0 + h)^T \varepsilon | X)$.

Using $\text{Var}(\varepsilon | X) = \sigma^2 I_n$, $\text{Var}(a^T y | X) = (a_0 + h)^T \sigma^2 I_n (a_0 + h) = \sigma^2 \|a_0 + h\|^2$.

Considering $a_0 = X(X^T X)^{-1}c \in \text{Col}(X)$ and $h \in \ker(X^T) = \text{Col}(X)^\perp$, $a_0^\top h = 0$.

$$\|a_0 + h\|^2 = \|a_0\|^2 + 2a_0^\top h + \|h\|^2 = \|a_0\|^2 + \|h\|^2.$$

Thus $\text{Var}(a^T y | X) = \text{Var}(a_0^T y | X) + \sigma^2 \|h\|^2$.

Since $\sigma^2 \|h\|^2 \geq 0$, OLS has variance no larger than any other linear unbiased estimator.

This proves Gauss-Markov for the scalar estimand $c^T \beta$. The proof is exactly a Pythagorean theorem. Any other unbiased estimator adds an orthogonal noise component $h^T y$ that does not help unbiasedness but does increase variance.

Exercise 2. Frisch-Waugh-Lovell (FWL)

Consider the two-block regression $y = X_1 \beta_1 + x_2 \beta_2 + \varepsilon$, where $y, x_2, \varepsilon \in \mathbb{R}^n$, $X_1 \in \mathbb{R}^{n \times p_1}$ has full column rank, and $\beta_2 \in \mathbb{R}$ is a scalar. Think of X_1 as the controls and x_2 as the variable of interest, and we investigate the effect of x_2 holding X_1 fixed. We will show that the OLS coefficient $\hat{\beta}_2$ from the joint regression equals a simple regression that uses only leftover variation after removing what X_1 can explain.

Define the hat and residual-maker matrices $H_1 = X_1(X_1^\top X_1)^{-1}X_1^\top$, $M_1 = I - H_1$. We know $H_1^\top = H_1 = H_1^2$, $H_1 X_1 = X_1$, $M_1^\top = M_1 = M_1^2$, $M_1 X_1 = 0$. $M_1 v$ is what remains of a vector v after subtracting its OLS fit on X_1 .

(i) Fix β_2 and minimize $\|y - X_1 \beta_1 - x_2 \beta_2\|_2^2$ over β_1 only. Show that the optimal residual (after choosing that best β_1) is $M_1(y - x_2 \beta_2)$. Conclude that the joint least-squares problem reduces to $\min_{\beta_2} \|M_1 y - M_1 x_2 \beta_2\|_2^2$.

(ii) Assume $M_1 x_2 \neq 0$. Differentiate the objective in (i) (or write the normal equation for a through-origin simple regression) and derive $\hat{\beta}_2 = \frac{x_2^\top M_1 y}{x_2^\top M_1 x_2}$.

(iii) Now suppose the true model is still $y = X_1 \beta_1 + x_2 \beta_2 + \varepsilon$ with $\mathbb{E}[\varepsilon | X_1, x_2] = 0$, but we fit the short regression of y on X_1 only: $\tilde{\beta}_1 = (X_1^\top X_1)^{-1} X_1^\top y$. Derive $\mathbb{E}[\tilde{\beta}_1 | X_1, x_2]$ and identify the omitted-variable bias term. When is that bias zero?

Solution 2

(i) For fixed β_2 , the target $y - x_2 \beta_2$ is known. Minimizing $\|(y - x_2 \beta_2) - X_1 \beta_1\|_2^2$ is OLS of $y - x_2 \beta_2$ on X_1 . The fitted vector in $\text{Col}(X_1)$ is $H_1(y - x_2 \beta_2)$, so the residual is $(y - x_2 \beta_2) - H_1(y - x_2 \beta_2) = M_1(y - x_2 \beta_2)$.

At this optimal β_1 , the joint criterion equals that residual norm. Linearity of M_1 gives $M_1(y - x_2 \beta_2) = M_1 y - M_1 x_2 \beta_2$, hence $\min_{\beta_2} \|M_1 y - M_1 x_2 \beta_2\|_2^2$.

Here $M_1 y$ is y after removing the linear effect of the controls X_1 ; $M_1 x_2$ is x_2 after removing the part already explained by X_1 .

(ii) $Q(\beta_2) = \|M_1 y\|_2^2 - 2\beta_2 (M_1 x_2)^\top (M_1 y) + \beta_2^2 \|M_1 x_2\|_2^2$.

Set $Q'(\beta_2) = 0$: $(M_1 x_2)^\top (M_1 x_2) \hat{\beta}_2 = (M_1 x_2)^\top (M_1 y)$.

Using $M_1^\top = M_1$ and $M_1^2 = M_1$, $(M_1 x_2)^\top (M_1 y) = x_2^\top M_1^\top M_1 y = x_2^\top M_1 y$,

and likewise $(M_1 x_2)^\top (M_1 x_2) = x_2^\top M_1 x_2$. Therefore

$$\hat{\beta}_2 = \frac{x_2^\top M_1 y}{x_2^\top M_1 x_2} = \frac{\tilde{x}_2^\top \tilde{y}}{\tilde{x}_2^\top \tilde{x}_2}, \quad \tilde{y} = M_1 y, \quad \tilde{x}_2 = M_1 x_2.$$

So the multiple-regression slope on x_2 is the simple through-origin slope of residualized y on residualized x_2 .

"Holding X_1 fixed" means only the part of x_2 orthogonal to $\text{Col}(X_1)$ is used to identify $\hat{\beta}_2$.

Comment. If $M_1 x_2 = 0$, then $x_2 \in \ker(M_1) = \text{Col}(X_1)$. x_2 is an exact linear combination of the controls. The denominator $x_2^\top M_1 x_2 = \|M_1 x_2\|_2^2 = 0$, so $\hat{\beta}_2$ is not identified. This is exact multicollinearity after residualizing, where no unique partial effect remains.

(iii) Substitute the true model into the short regression:

$$\tilde{\beta}_1 = (X_1^\top X_1)^{-1} X_1^\top (X_1 \beta_1 + x_2 \beta_2 + \varepsilon) = \beta_1 + (X_1^\top X_1)^{-1} X_1^\top x_2 \beta_2 + (X_1^\top X_1)^{-1} X_1^\top \varepsilon.$$

With $\mathbb{E}[\varepsilon \mid X_1, x_2] = 0$, $\mathbb{E}[\tilde{\beta}_1 \mid X_1, x_2] = \beta_1 + (X_1^\top X_1)^{-1} X_1^\top x_2 \beta_2$.

The omitted-variable bias is $(X_1^\top X_1)^{-1} X_1^\top x_2 \beta_2$. It vanishes if $\beta_2 = 0$ (omitted variable does not affect y) or $X_1^\top x_2 = 0$ (omitted variable is orthogonal to the included columns).

Exercise 3. Polynomial regression and overfitting

Observe pairs (x_i, y_i) for $i = 1, \dots, n$ with distinct x_i . For degree d , form the Vandermonde-type design

$$X_d = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^d \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^d \end{bmatrix} \in \mathbb{R}^{n \times (d+1)}$$

and the model $Y_i = \beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d + \varepsilon_i$. Write H_d for the orthogonal projector onto $\text{Col}(X_d)$, assume always well-defined.

When $d = n - 1$, X_{n-1} is invertible: the Vandermonde determinant is $\prod_{1 \leq i < j \leq n} (x_j - x_i) \neq 0$ when all x_i are distinct. For every y there is a unique β with $X_{n-1} \beta = y$, hence $\hat{y} = y$ and $SSE = 0$. The model interpolates any $y \in \mathbb{R}^n$, which means every pair (x_i, y_i) is learned, the start of overfit.

The target of polynomial regression here is to fit the mean $\mu = \mathbb{E}[y]$, not the noisy sample y itself. To discuss the bias-variance tradeoff, one can understand bias as systematic misspecification, how far $\mathbb{E}[\hat{y}]$ sits from μ , how much of μ lies outside $\text{Col}(X_d)$; and variance as how much $\hat{y}$ wobbles across noise draws ε , or roughly how large $\text{rank}(H_d)$ is).

(i) Can the training SSE increase when the degree goes from d to $d + 1$?

(ii) Assume $y = \mu + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma^2 I_n$. Using $e_d = (I - H_d)y$ and the quadratic-form identity $\mathbb{E}[\varepsilon^\top A \varepsilon] = \sigma^2 \text{tr}(A)$ for fixed symmetric A , derive

$$\mathbb{E}[\|(I - H_d)y\|_2^2] = \|(I - H_d)\mu\|_2^2 + \sigma^2(n - \text{rank}(H_d)).$$

Interpret the two terms on the right-hand side.

(iii) Discuss $d \geq n - 1$, and keep the assumptions of (ii). Let $y^{\text{new}} = \mu + \varepsilon^{\text{new}}$ on the same x -grid, with ε^{new} independent of ε , $\mathbb{E}[\varepsilon^{\text{new}}] = 0$, and $\text{Var}(\varepsilon^{\text{new}}) = \sigma^2 I_n$. Now $H_d = I_n$ and $\hat{y} = y = \mu + \varepsilon$.

- Derive $\mathbb{E}\|y^{\text{new}} - \hat{y}\|_2^2 = 2n\sigma^2$.

- Compare to the oracle predictor μ with $\mathbb{E}\|y^{\text{new}} - \mu\|_2^2 = n\sigma^2$, and conclude the effect of overfit as: interpolating OLS pays an extra $n\sigma^2$ of expected squared error equal to $\mathbb{E}\|\varepsilon\|_2^2$.

(iv) Keep $y = \mu + \varepsilon$ as in (ii), and write $\hat{y} = H_d y$ for a general degree d (not only $d \geq n - 1$).

- Show $\mathbb{E}[\hat{y}] = H_d \mu$, and derive

$$\mathbb{E}\|\hat{y} - \mu\|_2^2 = \|(H_d - I)\mu\|_2^2 + \sigma^2 \text{rank}(H_d).$$

The first term is squared bias. The second is total variance of $\hat{y}$.

- Using independence of ε^{new} as in (iii), show the prediction risk decomposes as

$$\mathbb{E}\|y^{\text{new}} - \hat{y}\|_2^2 = \|(H_d - I)\mu\|_2^2 + \sigma^2 \text{rank}(H_d) + n\sigma^2.$$

- Evaluate both extremes: (a) $\mu \in \text{Col}(X_d)$ but $H_d \neq I$; (b) $d \geq n - 1$. Explain the bias-variance tradeoff as d grows.

(v) Ridge regression mitigates overfit by modifying the OLS criterion by a penalty term, known as a regularization technique. For a fixed $\lambda > 0$ and design $X = X_d$, solve $\min_{\beta \in \mathbb{R}^{d+1}} \|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2$.

- By differentiating, derive the normal equation $(X^\top X + \lambda I)\hat{\beta}_\lambda = X^\top y$.
- Show that $X^\top X + \lambda I$ is invertible even if $d \geq n$ (so $X^\top X$ is singular).
- Let $\hat{y}_\lambda = X\hat{\beta}_\lambda$. Prove that $\hat{y}_\lambda = y$ forces $y = 0$. What does ridge avoid?

Solution 3

(i) Adding the column of x_i^{d+1} values enlarges the column space: $\text{Col}(X_d) \subseteq \text{Col}(X_{d+1})$. Training SSE is the squared distance from y to the model space, so $SSE_d = \min_{v \in \text{Col}(X_d)} \|y - v\|_2^2$, $SSE_{d+1} = \min_{v \in \text{Col}(X_{d+1})} \|y - v\|_2^2$. Minimizing over a larger set cannot increase the minimum, hence $SSE_{d+1} \leq SSE_d$.

(ii) Write $e_d = (I - H_d)y = (I - H_d)\mu + (I - H_d)\varepsilon$. Expand the squared norm:

$$\|e_d\|_2^2 = \|(I - H_d)\mu + (I - H_d)\varepsilon\|_2^2 = \|(I - H_d)\mu\|_2^2 + 2\mu^\top (I - H_d)^\top (I - H_d)\varepsilon + \|(I - H_d)\varepsilon\|_2^2.$$

Because H_d is an orthogonal projector, $H_d^\top = H_d = H_d^2$, hence $I - H_d$ is also symmetric and idempotent: $(I - H_d)^\top = I - H_d$ and $(I - H_d)^2 = I - H_d$. Therefore $(I - H_d)^\top (I - H_d) = I - H_d$, and

$$\|e_d\|_2^2 = \|(I - H_d)\mu\|_2^2 + 2\mu^\top (I - H_d)\varepsilon + \varepsilon^\top (I - H_d)\varepsilon.$$

Considering $\mathbb{E}[\varepsilon] = 0$ and that $I - H_d$ is nonrandom given the design, $\mathbb{E}[2\mu^\top (I - H_d)\varepsilon] = 2\mu^\top (I - H_d)\mathbb{E}[\varepsilon] = 0$.

Also, $\varepsilon^\top A \varepsilon = \text{tr}(A \varepsilon \varepsilon^\top) \implies \mathbb{E}[\varepsilon^\top A \varepsilon] = \text{tr}(A \mathbb{E}[\varepsilon \varepsilon^\top]) = \sigma^2 \text{tr}(A)$.

Apply with $A = I - H_d$: $\mathbb{E}[\varepsilon^\top (I - H_d)\varepsilon] = \sigma^2 \text{tr}(I - H_d) = \sigma^2(n - \text{tr}(H_d))$.

For an orthogonal projector, $\text{tr}(H_d) = \text{rank}(H_d)$, so

$$\mathbb{E}[\|(I - H_d)y\|_2^2] = \|(I - H_d)\mu\|_2^2 + \sigma^2(n - \text{rank}(H_d)).$$

The first term is approximation error (how much of μ lies outside the degree- d column space). The second is residual noise after projection, with effective dimension $n - \text{rank}(H_d)$.

(iii) Write $y^{\text{new}} - \hat{y} = (\mu + \varepsilon^{\text{new}}) - (\mu + \varepsilon) = \varepsilon^{\text{new}} - \varepsilon$.

Square: $\|y^{\text{new}} - \hat{y}\|_2^2 = \|\varepsilon^{\text{new}}\|_2^2 + \|\varepsilon\|_2^2 - 2(\varepsilon^{\text{new}})^\top \varepsilon$.

Take expectations. Independence of ε^{new} and ε together with $\mathbb{E}[\varepsilon^{\text{new}}] = 0$ kills the cross term: $\mathbb{E}[(\varepsilon^{\text{new}})^\top \varepsilon] = \mathbb{E}[\varepsilon^{\text{new}}]^\top \mathbb{E}[\varepsilon] = 0$.

Also $\mathbb{E}\|\varepsilon\|_2^2 = \text{tr}(\text{Var}(\varepsilon)) = n\sigma^2$, and likewise $\mathbb{E}\|\varepsilon^{\text{new}}\|_2^2 = n\sigma^2$.

Hence $\mathbb{E}\|y^{\text{new}} - \hat{y}\|_2^2 = n\sigma^2 + n\sigma^2 = 2n\sigma^2$.

For μ as a predictor, $y^{\text{new}} - \mu = \varepsilon^{\text{new}}$, $\mathbb{E}\|y^{\text{new}} - \mu\|_2^2 = \mathbb{E}\|\varepsilon^{\text{new}}\|_2^2 = n\sigma^2$.

The excess risk of the interpolating fit over this oracle is $\mathbb{E}\|y^{\text{new}} - \hat{y}\|_2^2 - \mathbb{E}\|y^{\text{new}} - \mu\|_2^2 = 2n\sigma^2 - n\sigma^2 = n\sigma^2 = \mathbb{E}\|\varepsilon\|_2^2$,

which is also $\mathbb{E}\|\hat{y} - \mu\|_2^2$. Training SSE looks perfect because $\hat{y}$ swallowed the entire noise vector ε . Nevertheless, on new data that memorized ε is useless and costs an extra $n\sigma^2$ in expected squared error, exactly the size of the noise that was fitted.

(iv) Define the ridge objective (as in the question)

$$f(\beta) = \|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2, \quad X = X_d, \quad \lambda > 0.$$

Expand $\|y - X\beta\|_2^2 = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X\beta$, so

$$f(\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X\beta + \lambda\beta^\top \beta.$$

Differentiate with respect to the vector β , using $\nabla(\beta^\top A\beta) = (A + A^\top)\beta$ for square A , here $A = X^\top X$ and $A = \lambda I$ are symmetric:

$$\nabla f(\beta) = -2X^\top y + 2X^\top X\beta + 2\lambda\beta.$$

A critical point $\hat{\beta}_\lambda$ satisfies $\nabla f(\hat{\beta}_\lambda) = 0$, i.e.

$$(X^\top X + \lambda I)\hat{\beta}_\lambda = X^\top y.$$

The Hessian $2(X^\top X + \lambda I)$ will be shown positive definite next, so this critical point is the unique global minimizer. For every $v \in \mathbb{R}^{d+1}$ and $v \neq 0$,

$$v^\top (X^\top X + \lambda I)v = v^\top X^\top Xv + \lambda v^\top v = \|Xv\|_2^2 + \lambda\|v\|_2^2 \geq \lambda\|v\|_2^2 > 0.$$

Therefore $X^\top X + \lambda I$ is (symmetric) positive definite, hence invertible, and

$$\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$$

is well-defined even in the fat- X regime where OLS has no unique solution.

Let $\hat{y}_\lambda = X\hat{\beta}_\lambda$ and suppose $\hat{y}_\lambda = y$. Then $X^\top y = X^\top X\hat{\beta}_\lambda$. Substitute into the normal equation:

$$(X^\top X + \lambda I)\hat{\beta}_\lambda = X^\top y = X^\top X\hat{\beta}_\lambda, \quad X^\top X\hat{\beta}_\lambda + \lambda\hat{\beta}_\lambda = X^\top X\hat{\beta}_\lambda, \quad \lambda\hat{\beta}_\lambda = 0$$

Since $\lambda > 0$, we get $\hat{\beta}_\lambda = 0$, hence $y = X\hat{\beta}_\lambda = 0$. Contrapositively: if $y \neq 0$, then $\hat{y}_\lambda \neq y$.

Ridge therefore cannot reproduce the interpolating fit $\hat{y} = y$ from (iii) that paid the extra $n\sigma^2$ prediction cost because the $\lambda\|\beta\|_2^2$ term forces a compromise between fitting y and keeping $\|\beta\|_2$ small.